

REETUNJAI DUTTA

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PROFESSIONAL SUMMARY

Aerospace Engineering student (B.Tech 2023–27, CGPA: 7.9) with hands-on experience in satellite systems engineering, CAD design, structural simulation, and embedded systems. Contributed to a 1U CubeSat as Systems Engineer Intern and published research at IAC 2025 (Sydney, Australia). Proficient in Ansys FEA/CFD, Solidworks, ArduPilot, Python, and Arduino. Silver Medalist in Drone Race and runner-up in Robotics Race at Amity University. Seeking to apply technical expertise and space-sector passion at Dhruva Space.

TECHNICAL SKILLS

CAD & Design: Solidworks, Autodesk Fusion 360, OnShape, Shapr3D

Simulation & Analysis: Ansys FEA, Ansys CFD, CST Studio Suite, Finite Element Analysis

Programming: Python, C Embedded (Beginner), Arduino IDE

Hardware & Embedded: Arduino UNO, ESP32, ESP8266, Raspberry Pi

Aerospace Systems: Satellite Systems (1U CubeSat), ArduPilot, BLDC Motors/ESC, Rocker-Bogie Suspension

Fabrication & Other: 3D Printing (PLA), Rapid Prototyping, Robotics & IoT Systems, Research & Technical Writing

WORK EXPERIENCE

IoT & Robotics Intern — ROBOTICS SIR 2026
Conditional Internship Offer (PPO)

- Working on CAD prototyping and IoT projects for their Education KIT product line.
- Received Pre-Placement Offer (PPO) based on internship performance.

Technologies Used: CAD, IoT, Solidworks, Prototyping

Faculty — IoT & Robotics — TEAM MECHANICS 2025

- Working as faculty to teach students about IoT systems, embedded electronics, and Robotics fundamentals.

Technologies Used: Arduino, ESP32, IoT Systems, Embedded C

Systems Engineer Intern — STAR (Space Technology and Aeronautical Rocketry) 2024

- Contributing to the mechanical design, structural simulation, and integration of a 1U CubeSat.
- Performing FEA simulations in Ansys to validate structural integrity of satellite components.
- Supporting electronics integration and learning subsystem interfaces for the CubeSat platform.
- Collaborating within a multidisciplinary engineering team following satellite development workflows.

Technologies Used: Solidworks, Ansys FEA, Ansys CFD, Arduino, Systems Engineering

PROJECTS

IAC 2025 Research Paper — Lead Author 2025

- Led a research team; authored paper selected for the International Astronautical Congress 2025, Sydney, Australia (DOI: 083102-0052).
- Paper: "Fourth Space: Innovating Public Engagement to Unlock Inclusive Space Business Models." Demonstrated space-sector research and team leadership.

3D-Printed RC Aircraft & RC Boat 2024

- Designed and 3D-printed a functional RC aircraft using PLA; applied aerodynamic design and rapid prototyping principles.
- Built a BLDC-powered RC racing boat and a wireless robo car using ESP8266 and motor driver modules.

Technologies Used: Solidworks, 3D Printing, BLDC, ESP8266, Arduino IDE

1U CubeSat — Mechanical Design & Simulation 2024

- Designed and simulated CubeSat structural components; validated frame integrity against launch and vibration load requirements.
- Applied Ansys FEA for stress and deformation analysis; iterated designs for weight reduction and structural compliance.
- Gained hands-on experience in satellite subsystem integration and space hardware documentation.

Racing Drone — ArduPilot Flight Controller 2023

- Designed and built a racing drone; integrated BLDC motors, ESC, and telemetry; tuned PID parameters for stable flight.
- Won Silver Medal at Inter-University Drone Race Competition, Amity University 2024.

Technologies Used: ArduPilot, BLDC Motors, ESC, PID Tuning, Arduino

ASTRIX Rover — CAD Design 2024 – Present

- Designing a planetary rover with rocker-bogie suspension in Solidworks/Fusion 360; applying mechanical design principles for terrain adaptability.

EDUCATION

Bachelor of Technology — Aerospace Engineering 2023 – 2027
Amity University Haryana CGPA: 7.9

CERTIFICATIONS & ACHIEVEMENTS

Selected for IP – International Astronautical Congress (IAC 2025), Sydney, Australia | DOI: 083102-0052

Silver Medalist – Drone Race Competition, Inter-University Fest, Amity University 2024

2nd Prize – Robotics Race, Amifest 2026, Amity University Haryana

Certificate of Participation – IoT Mobile Robotics with Arduino, TRYST-IIT Delhi (March 2024)

Certificate of Participation – ISRO AI Immersion Challenge, IIIT Hyderabad & NRSC-ISRO (August 2024)

Certification – Aerospace Engineering: Rocket Science and Engineering (Udemy)

Certification – Drone Systems and Control (SWAYAM Portal)